CHARGER M1 / SOLAR + CELL PADS / S1

M1
TP4056 module
CHARGE_5V 1
CHARGE_RETURN 2
PROTECTED_POS 3
GND_LOAD 4
PACK_POS 5
PACK_NEG 6

JPV1
PV IN + / −
CHARGE_5V 1
CHARGE_RETURN 2

JCELL1
CELL B+ / B−
PACK_POS 1
PACK_NEG 2

JS1
S1 OUT+ / SYS
PROTECTED_POS 1
VBAT_SYS 2

CONTROLLER + ISP / RAW NC

JM2L1
M2 LEFT / serial end top
UART_TX 1
2
AVR_RESET 3
GND_LOAD 4
TEST 5
6
7
8
9
LDR_POWER 10
11
PWM 12

JM2R1
M2 RIGHT / serial end top
GND_LOAD 1
2
AVR_RESET 3
VBAT_SYS 4
5
6
LDR_SENSE 7
8
ISP_SCK 9
ISP_MISO 10
ISP_MOSI 11
12

JISP1
2x3 male header 2.54 mm
ISP_MISO 1
2 VBAT_SYS
ISP_SCK 3
4 ISP_MOSI
AVR_RESET 5
6 GND_LOAD

OPTIONAL D1 MINI SOCKETS

JM3L1
M3 LEFT / antenna end top
ESP_RESET 1
ADC_BATT 2
ESP_D0 3
4
UART_RX_INVERTED 5
6
7
ESP_3V3 8

JM3R1
M3 RIGHT / antenna end top
1
2
3
4
5
6
GND_LOAD 7
8

C3
100u / 10V
+
VBAT_SYS
GND_LOAD

C4
100n
VBAT_SYS
GND_LOAD

UART INVERTER / BATTERY SENSE

R8
47k
UART_TX
UART_BASE

R9
100k
UART_BASE
GND_LOAD

Q2
2N3904 / EBC
M1
UART_BASE 2
UART_RX_INVERTED
GND_LOAD

R10
10k
ESP_3V3
UART_RX_INVERTED

LIGHTS + REMOTE SENSOR / BUTTON

R3
56R / 0.25W
PWM
LED_POS
LED_POS

JLED1
LED + / −
LED_POS 1
GND_LOAD 2

JLDR1
LDR D7 / A1
LDR_POWER 1
LDR_SENSE 2

R4
22k
LDR_SENSE
LDR_TRIM

R5
220k
SENSE_FEED
VBAT_SYS

JP3
ADC link
1
SENSE_FEED 2
ADC_BATT

JP2
D0 wake link
ESP_D0 1
ESP_RESET 2

RV1
50k 3296W
LDR_TRIM
2 GND_LOAD
GND_LOAD

EXTERNAL U3 / REMOVABLE LINKS

SW1
6x6 tactile
TEST
GND_LOAD

JU3IN1
U3 VIN + / −
VBAT_SYS 1
GND_LOAD 2

JU3OUT1
U3 VOUT + / −
REG_3V3 1
GND_LOAD 2

C2
470u / 10V
+
VBAT_SYS
GND_LOAD

JP1
BAT power link
REG_3V3 1
ESP_3V3 2

U0 solar buck, U3 buck-boost, cells and S1 are external. B− is its own net: never joined to GND_LOAD. No fuses. All returns = M1 OUT−.

Tier 0: omit telemetry. Tier 1: JP1 OPEN, USB. Tier 2: JP1 + JP2 closed, NO USB. JP3 closed only with powered M3.

R3 56R (was 47R in Rev B): about 19 mA peak at 4.2 V, still to confirm on the bench. No LED−cable short protection.

RV1 50k rheostat + R4 22k = LDR lower leg 22k−72k; mid-travel matches the Rev B 47k. SW1 is on the board.

No fuses; U0, U3 external

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Title: Solar Lights Lite − carrier with charger

Size: A3	Date: 28−Oct−2026	**Rev: C / DRAFT**
KiCad E.D.A. 10.0.6		Id: 1/1